



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO AVC ASSOCIATES

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

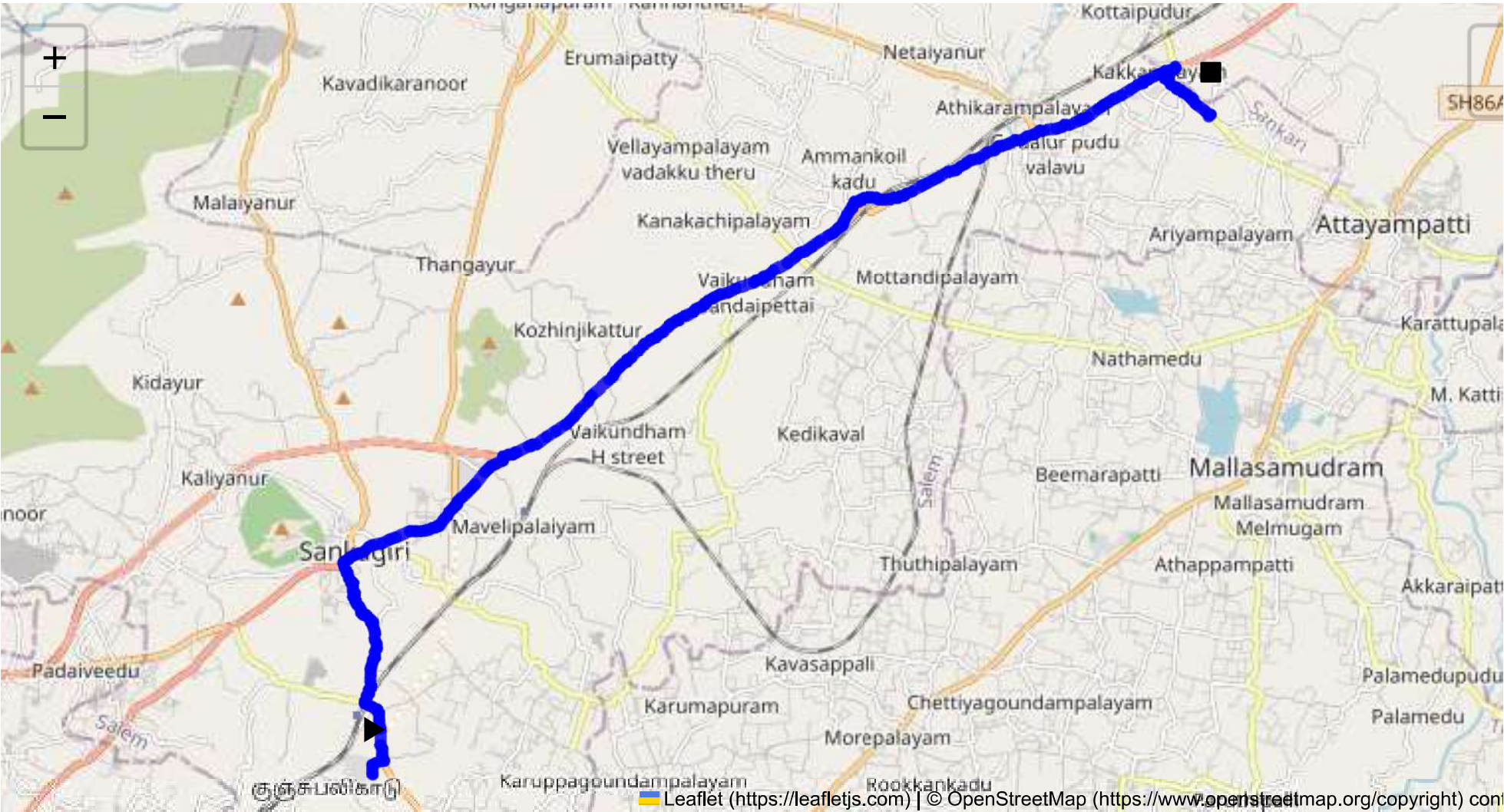
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 24.84 km
Estimated Duration: 0.6 hours
Adjusted Duration (Heavy Vehicle): 0.7 hours
Start: (11.4381, 77.8734)
End: (11.550074, 78.018722)



Welcome to the Journey Risk Management Study

Route Overview

The route from CVQF+23W, Sangagiri to H229+2CX, Attayampatti - Kakapalayam Rd, Salem, spans approximately 24.84 kilometers and mostly involves regional roads that are used for both local and through traffic. While there are no highways involved, the route includes intersections, potential small town crossings, and rural stretches.

Weather Conditions

Typical Weather: Tamil Nadu generally experiences a tropical climate with hot summers, a monsoon season, and mild winters.

- **Potential Hazards:** During monsoon (June to September), heavy rainfall can lead to slippery roads, water-logging, and reduced visibility. Summers (March to May) can cause roads to heat up, leading to potential tire issues.

Traffic Patterns

- **Peak Hours:** Morning (8:00 AM - 10:00 AM) and evening (4:00 PM - 7:00 PM) can witness increased traffic due to local commuter patterns.
- **Congestion-Prone Areas:** Expect congestion near town centers and major intersection points, especially during peak hours.

Road Quality and Infrastructure

- **Assessment:** The route comprises mostly rural roads, which may have varying road surface quality. Potholes and uneven surfaces are possible, particularly after the monsoon.
- **Infrastructure:** Pay attention at narrow bridges and sudden road narrowing in village areas.

Suggestions for Alternative Routes in Emergencies

- **Alternative Route:** Depending on the point of emergency, taking NH79 might be advisable as it offers quicker access to emergency services and better infrastructure.

Local Regulations Affecting Hazardous Material Transport

- **Regulations:** Ensure compliance with Tamil Nadu’s regulations for transporting hazardous materials, including proper labeling, secure loading, and adherence to notified routes.

Historical Incidents

- **Previous Incidents:** There have been few reported incidents of heavy vehicle accidents; however, care is advised during inclement weather or in low-visibility conditions.

Environmental Considerations

- **Sensitive Areas:** Crossing through village areas demands slower speeds and noise regulation adherence. Be mindful of local fauna and flora, particularly in rural stretches.

Communication Coverage

- **Coverage Analysis:** Most of the route has good mobile network coverage, but some rural sections may have short dead zones or poor signal strength.

Emergency Response Times

- Estimated Times:** Response times can vary; the proximity to Salem city should ensure quicker response in areas closer to city limits but expect delays in remote or rural sections.

Summary of Risk Assessment

- Overall Risk:** The route poses moderate risk primarily due to road quality and traffic congestion during peak hours. Weather conditions can further complicate matters, but adherence to safety regulations and preparation for alternative routes can mitigate many risks. Familiarity with local regulations and timely information on weather and traffic conditions will help optimize the safety of the journey.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Turn	High	11.43811, 77.87348	15 KM/Hr
1	Turn	Medium	11.43956, 77.87340	30 KM/Hr
2	Turn	Medium	11.43971, 77.87348	30 KM/Hr
3	Turn	High	11.44029, 77.87544	15 KM/Hr
4	Turn	High	11.45007, 77.87201	15 KM/Hr
5	Turn	High	11.47407, 77.86839	15 KM/Hr
6	Turn	High	11.48369, 77.88781	15 KM/Hr
7	Turn	Medium	11.48370, 77.88787	30 KM/Hr
8	Turn	Medium	11.49149, 77.89601	30 KM/Hr
9	Turn	High	11.49224, 77.89601	15 KM/Hr
10	Turn	Medium	11.55622, 78.00774	30 KM/Hr
11	Blind Spot	Blind Spot	11.55839, 78.01288	10 KM/Hr
12	Turn	High	11.55800, 78.01305	15 KM/Hr
13	Blind Spot	Blind Spot	11.55707, 78.01094	10 KM/Hr
14	Turn	High	11.55027, 78.01896	15 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	police	Police Quarters	11.4741243, 77.8673495	30 km/h	Medium
1	hospital	Kathir Eye Hospital	11.4740976, 77.8681591	30 km/h	Medium
2	hospital	Thilagam Hospital	11.4753573, 77.8685829	30 km/h	Medium
3	clinic	Meyyazhagan Clinic	11.4752254, 77.8685671	30 km/h	Medium
4	hospital	Dr. Jaganathan Hospital	11.4752717, 77.8709312	30 km/h	Medium
6	hospital	Sankari Hospital	11.4777496, 77.873892	30 km/h	Medium
8	clinic	Dr. Dhanasekar Clinic	11.4793113, 77.8829247	30 km/h	Medium
9	hospital	Sri Vellaiyan Hospital	11.4798876, 77.8831394	30 km/h	Medium
11	hospital	C. N. Hospital	11.55724, 78.0096	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
5	school	Goverment Boys Higher Secondary School	11.477659, 77.8659632	30 km/h	Medium
7	marketplace	Sunday Market	11.4796502, 77.8708458	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Turn
Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44029, 77.87544



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45007, 77.87201



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.47407, 77.86839



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.48369, 77.88781



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.48370, 77.88787



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.49149, 77.89601



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49224, 77.89601



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.55622, 78.00774



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.55800, 78.01305



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.55707, 78.01094



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.55027, 78.01896

